MRSIPrep report: sub-01 ses-01

Inputs

BIDS directory: `/data`

Output directory: `/data/derivatives/mrsiprep`

Processing mode: `parc-con`

MRSinMRS

Parameter	Value	Unit
AcquisitionDurationMS	389	ms
Averages	1	
AveragingMode	N.A.	
CoilElements	HEA;HEP	
CombinedMetabolites	tNAA (NAA+NAAG), tCr (Cr+PCr), Cho (GPC+PCh), Ins (mI), Glx (Glu+Gln)	
CompressedSensingAccelerationFactor	2.5	
DataProcessing	Low-rank + TGV reconstruction for ECCENTRIC; lipid/water removal	
DeltaFrequencyPPM	0.0	ppm
Dimension	3D	
FIDPointsVectorSize	512	
FOV	220 x 220 x 130	mm
FlipAngle	45.0	°
MagneticFieldStrength	3.0	T
MatrixSize	44 x 44 x 25	
Measurements	1	
MetaboliteBasisSetLCModel	NAA, NAAG, Cr, PCr, GPC, PCh, mI, sI, Glu, Gln, Lac, GABA, GSH, Tau, Asp, Ala	
Orientation	Transverse	
PhaseEncoding	Elliptical	
PreparationScans	4	
QualityMetrics	SNR, CRLB, FWHM	
Quantification	LCModel	
RFCoils	32Rx ch 1H head coil	
RemoveOversampling	N.A.	
RepetitionTime	0.457	s
ResolutionMM3	5 x 5 x 5.2	mm
RotationDeg	-0.79	°
SaturationBands	1 band, 20mm thickness	

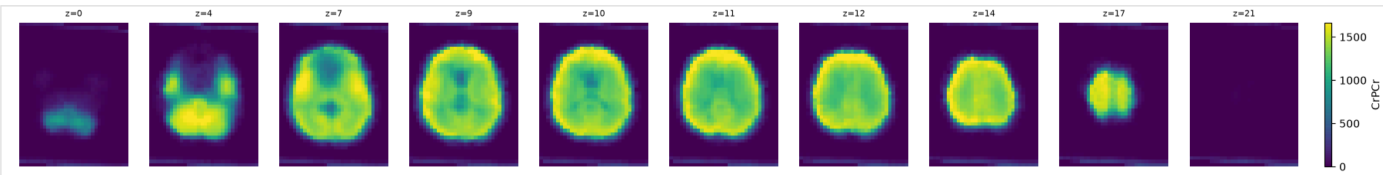
Scanner	3T Magnetom Prisma Fit (Siemens)	
Sequence	3D ECCENTRIC 1H-FID-MRSI	
ShimMode	Advanced	
SlabThicknessMM	95	mm
Slabs	1	
SpectralBandwidthHz	1320	Hz
SpectralSuppr	None	
TE	0.78	s
WaterReferenceFIDPoints	512	
WaterReferenceFlipAngle	45.0	°
WaterReferenceResolutionMM3	10.0 x 10.0 x 10.0	mm
WaterReferenceTE	0.72	s
WaterReferenceTR	0.46	s
WaterSuppr	WET water suppr.	
WaterSupprBWHz	80	Hz

MRSI Raw QC

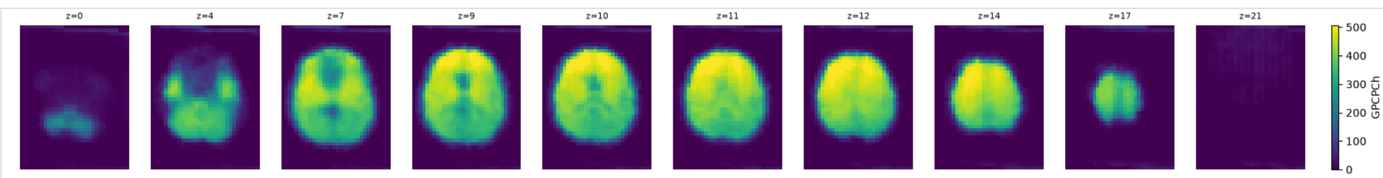
metabolite	n_total_voxels	mean_snr	median_snr	mean_linewidth	median_linewidth	mean_crlb	median_crlb
CrPCr	20738	14.257129	15.0	0.037799	0.036	6.431937	5.0
GluGln	20738	14.345459	15.0	0.037970	0.036	11.069957	11.0
GPCPCh	20738	14.228251	15.0	0.037878	0.036	6.853189	6.0
NAANAAG	20738	14.135942	15.0	0.038052	0.036	4.933007	4.0
Ins	20738	14.465190	15.0	0.037318	0.033	8.926435	8.0

Raw metabolite maps (pre-pipeline)

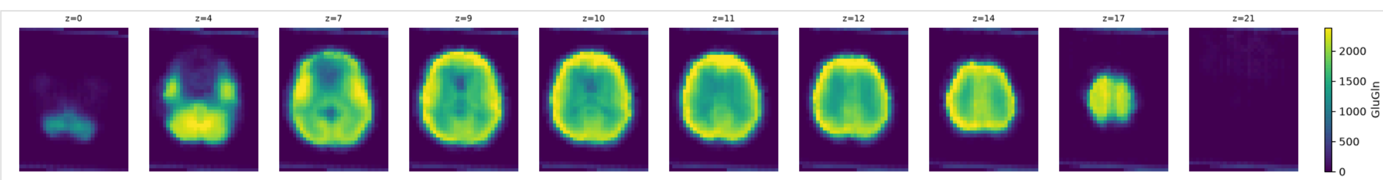
Metabolite: CrPCr



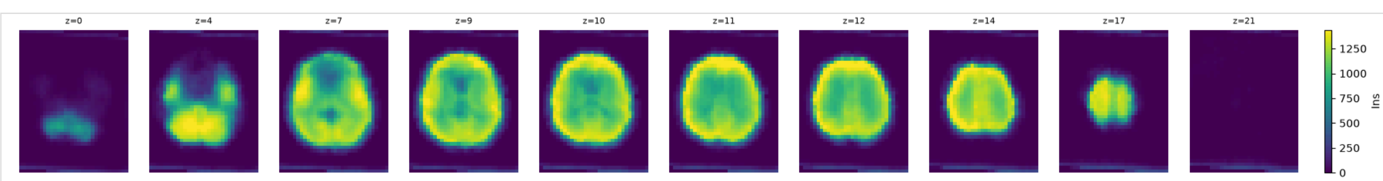
Metabolite: GPCPCh



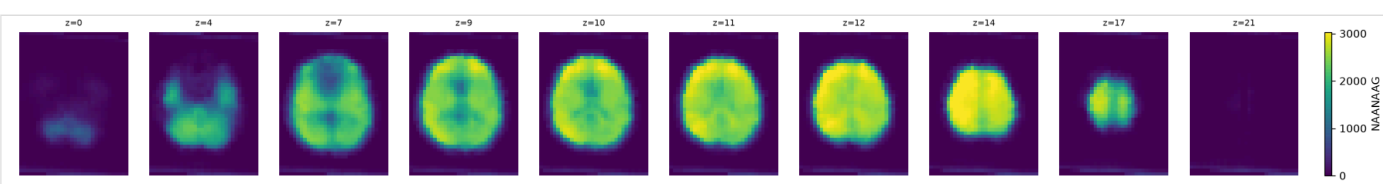
Metabolite: GluGln



Metabolite: Ins

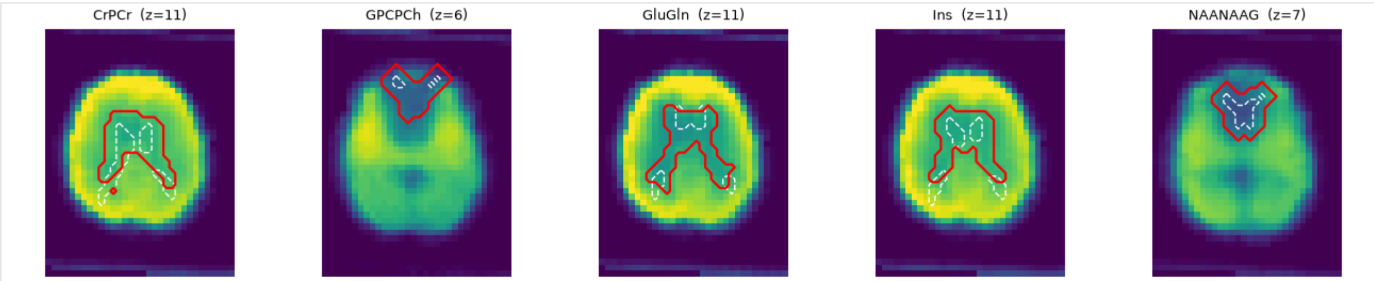


Metabolite: NAANAAG



Ventricle visibility (pre-coregistration)

Native-MRSI-space ventricle visibility, before any T1w coregistration: a cheap MNI-prior placement (dashed white) and the resulting data-driven detection (red) on each metabolite's own best-contrast slice. A clean, anatomically plausible outline indicates well-resolved ventricles; a ragged, offset, or absent one is worth a closer look before trusting downstream registration.



Preproc

Processing parameters

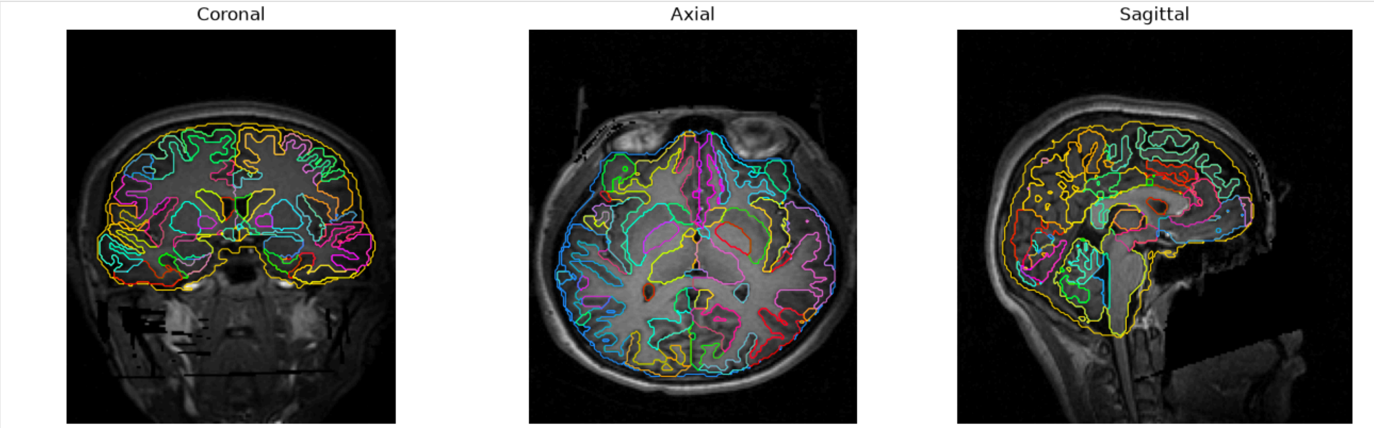
Parameter	Value
Processing mode	parc-con
Tissue backend	synthseg-fast
Registration backend	ants
Registration T1w target	brain-csf
Normalization	simple
Output spaces	MNI152NLin2009cAsym
Parcellation mode	mni
Atlas	chimera-LFMIHIFIS_scale3
SNR min	4.0
Linewidth max	0.1
CRLB max	20.0
Biharmonic spike filtering	True
Spike percentile	99.0
PVC disabled	False
T1 saturation correction	none
Connectivity	True
nproc x nthreads	1 x 16

Pipeline stages

`Validate inputs` → `Tissue segmentation` → `Anatomical prep` → `MRSI preprocessing` → `Registration` → `Tissue probability maps` → `Tissue QC` → `Partial volume correction` → `Resampling` → `Leakage QC` → `SynthSeg parcellation QC` → `Parcellation` → `Regional extraction` → `Connectivity` → `Metabolite profiles` → `Reports`

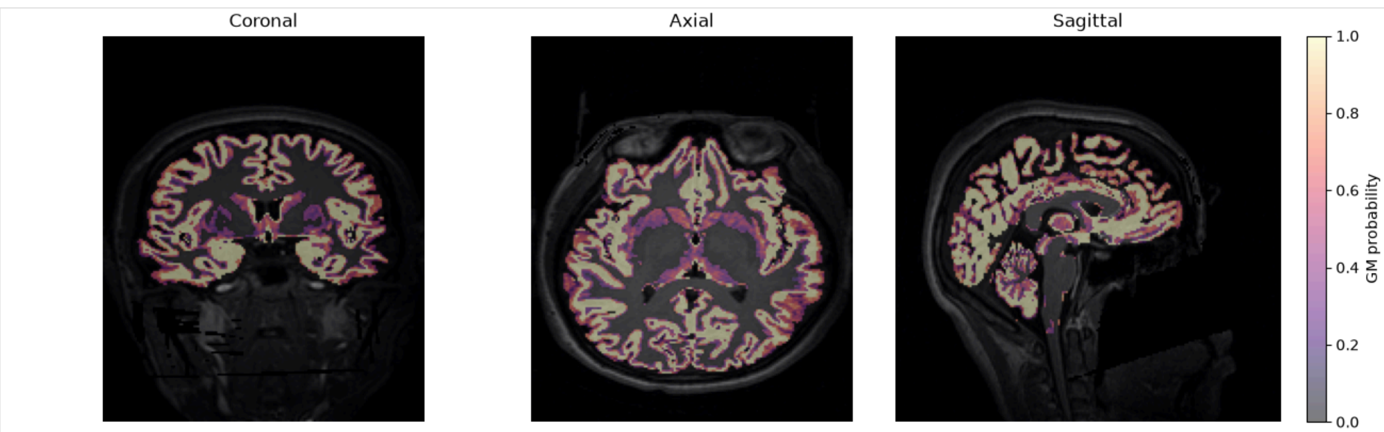
Anatomical

Tissue label outlines

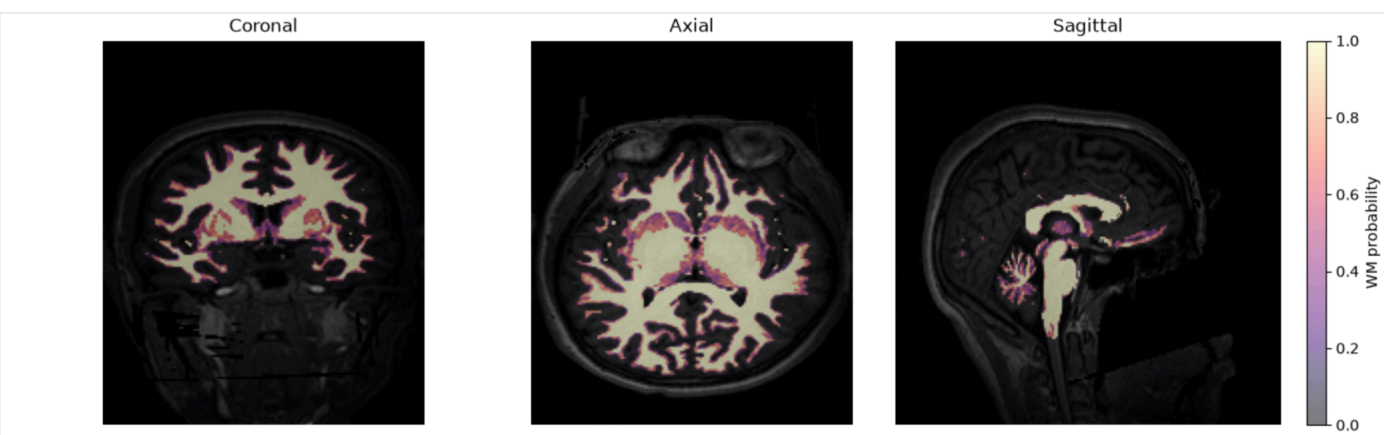


Tissue probability maps

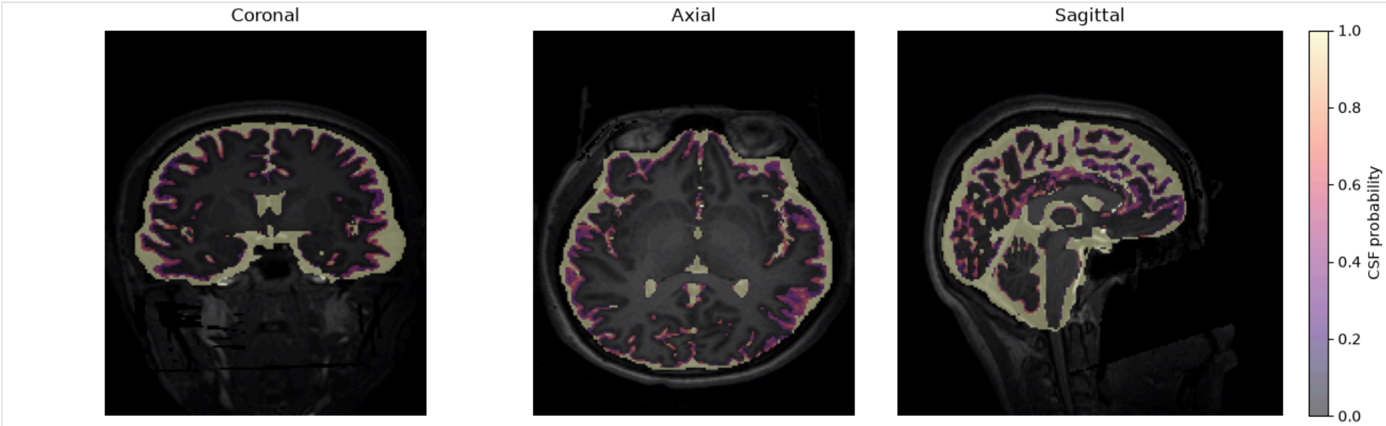
GM



WM



CSF


Coronal
Axial
Sagittal
1.0
0.8
0.6
0.4
0.2
0.0
CSF probability

Spike filter

Filtering summary

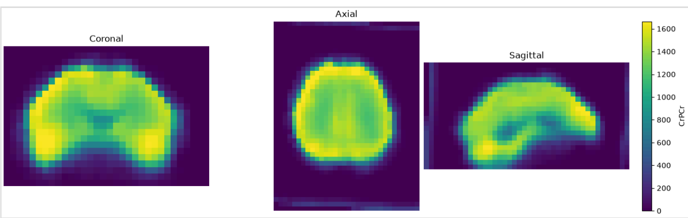
Slices centered on the centroid of all voxels repaired by spike/missing-voxel filtering, across all metabolites.

Metabolite	Spike voxels detected
CrPCr	7
GluGln	25
GPCPCh	0
NAANAAG	0
Ins	20

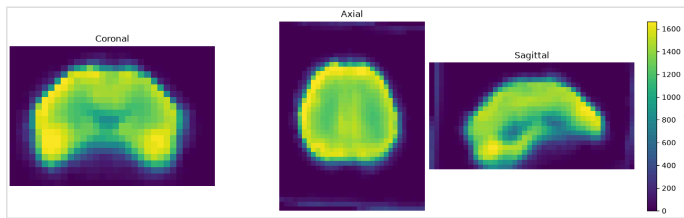
Metabolite: CrPCr

Spike voxels detected: 7

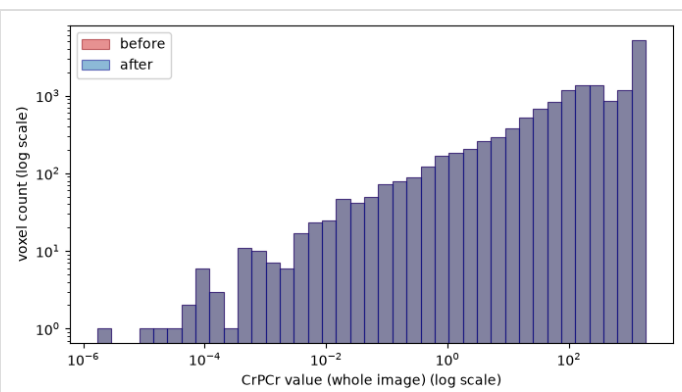
Before



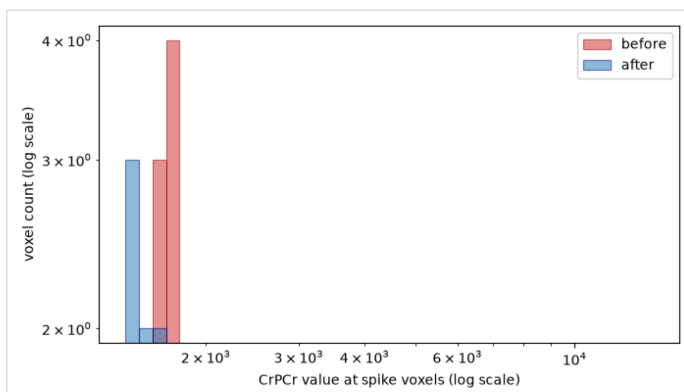
After



Whole image



Spike voxels only

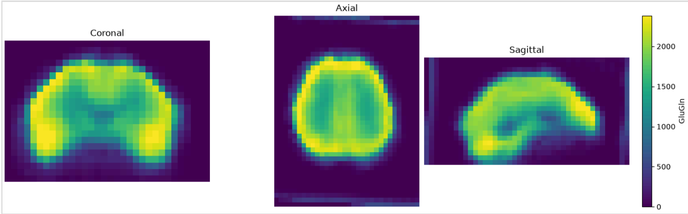


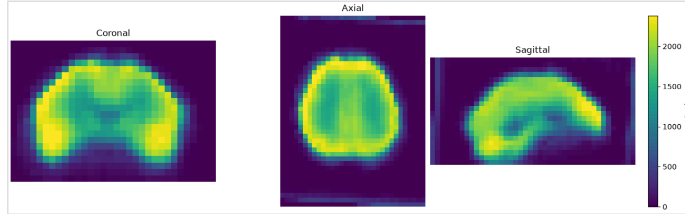
Metabolite: GluGln

Spike voxels detected: 25

Before

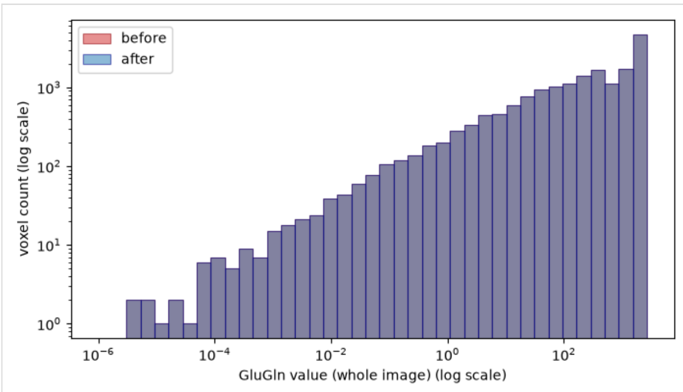
After

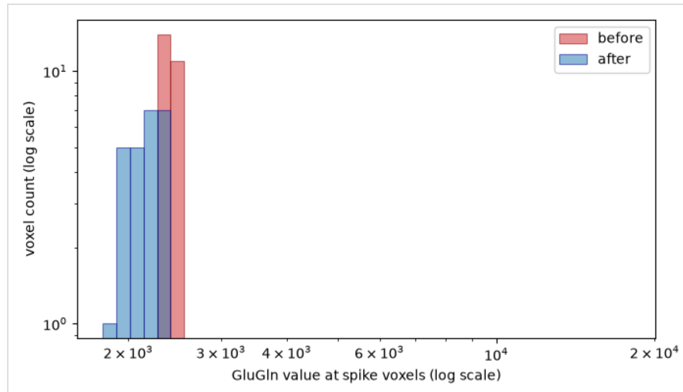




Whole image

Spike voxels only



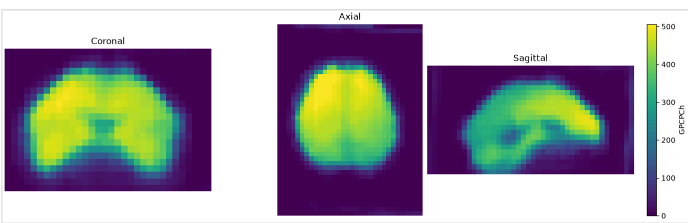


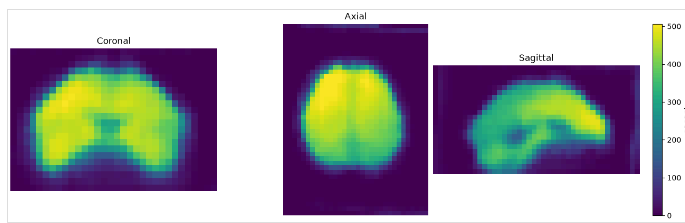
Metabolite: GPCPCh

Spike voxels detected: 0

Before

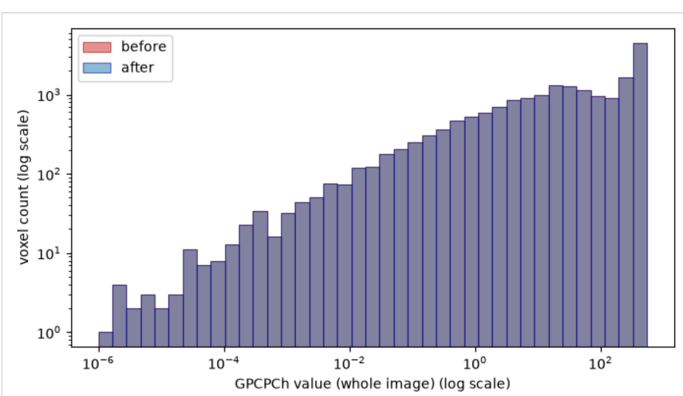
After

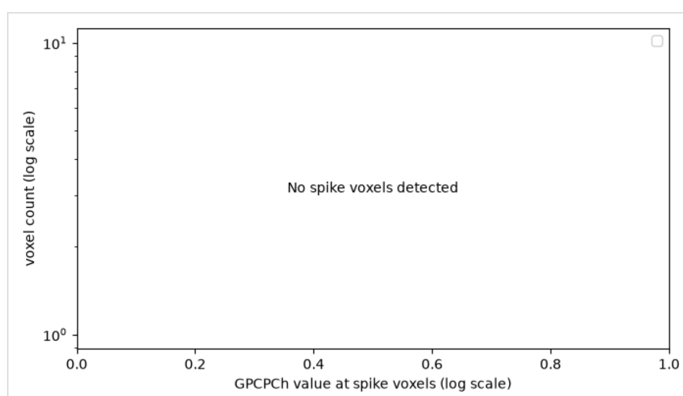




Whole image

Spike voxels only



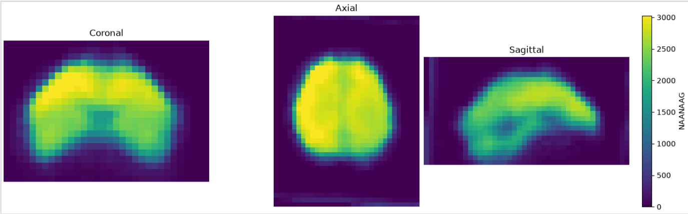


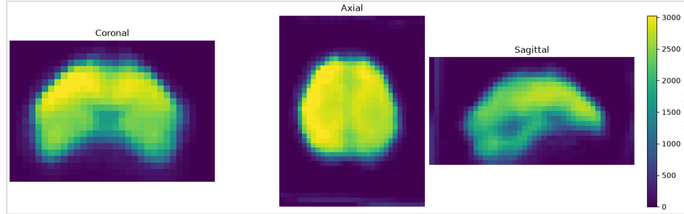
Metabolite: NAANAAG

Spike voxels detected: 0

Before

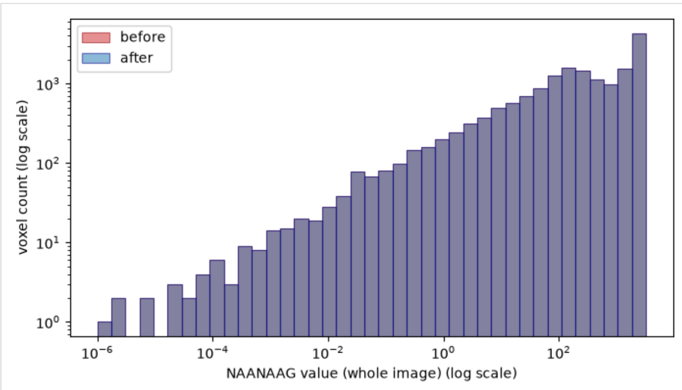
After

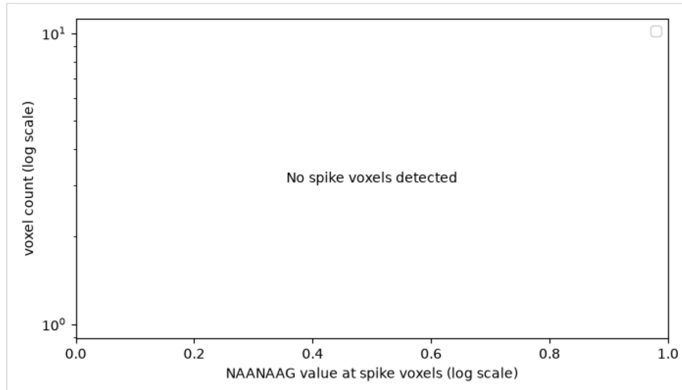




Whole image

Spike voxels only



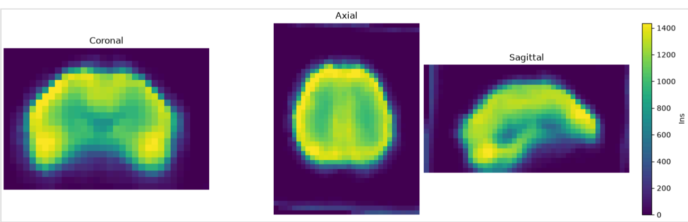


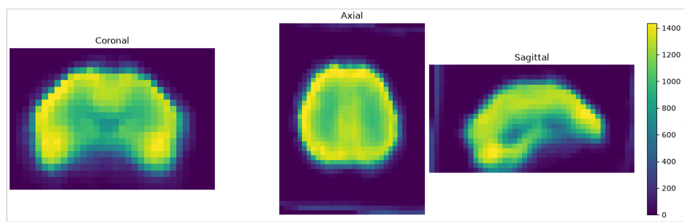
Metabolite: Ins

Spike voxels detected: 20

Before

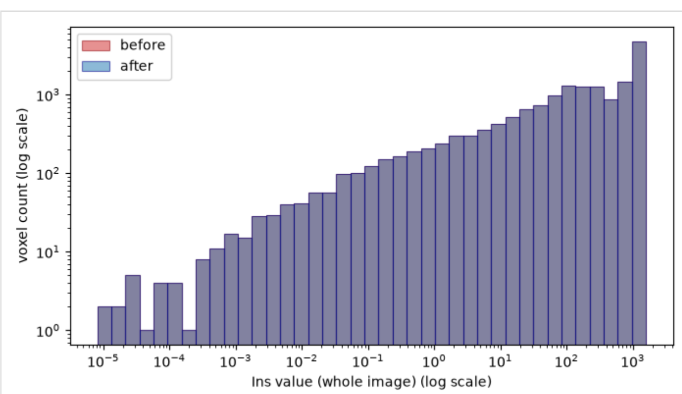
After

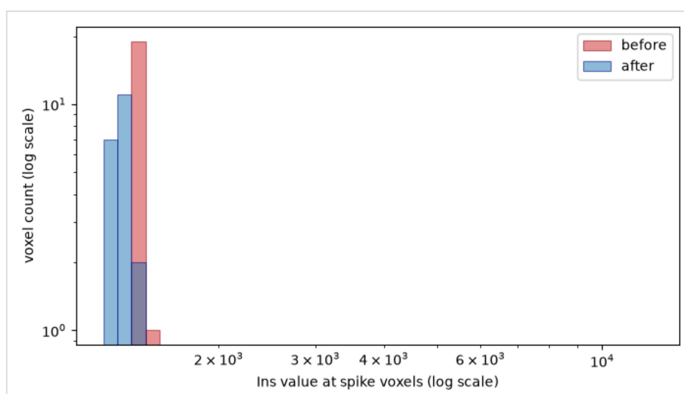




Whole image

Spike voxels only

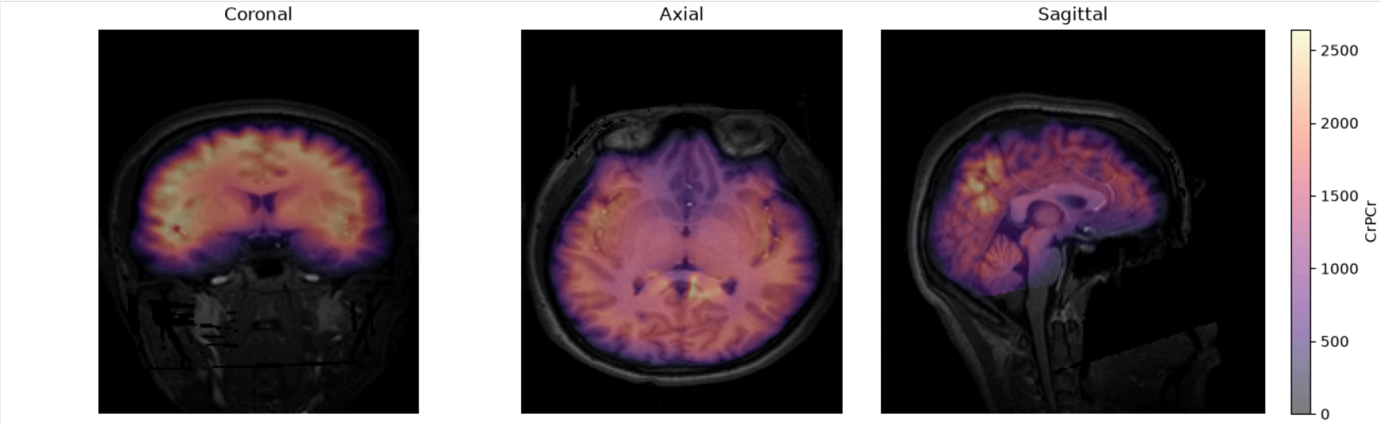




T1w-space alignment

T1w-space alignment

Reference metabolite map (`CrPCr`) overlaid on raw T1w.



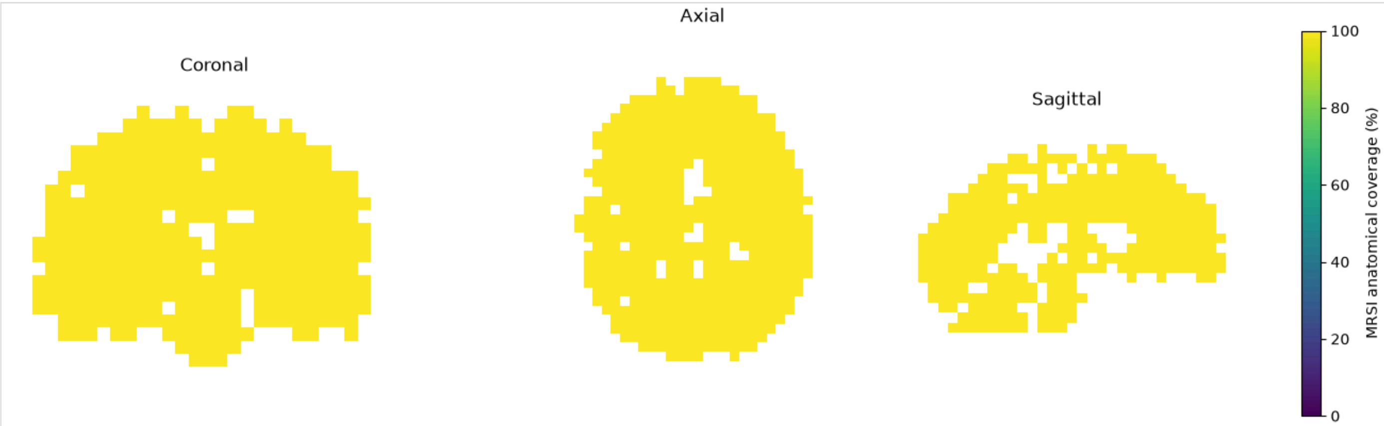
Signal-weighted leakage

Not computed: T1w-space leakage requires the full T1w-space resampled metabolite maps (`--output-mrsi-t1w`), which weren't requested for this run. The alignment image above uses a separate, lightweight report-only resampling and doesn't imply this data is available.

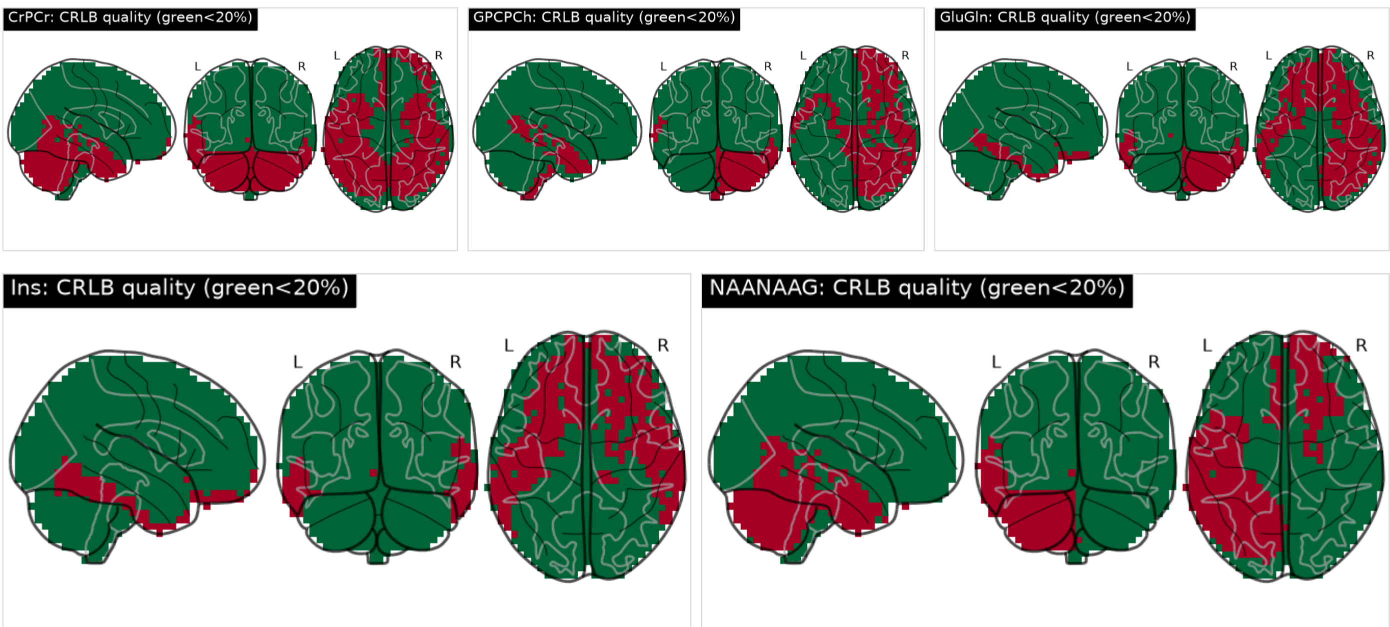
Coverage & Alignment

Mean anatomical MRSI coverage: **100.0%**; mean parcel CRLB: **18.66**.

MRSI anatomical coverage



Parcelwise CRLB quality (green reliable / red unreliable), 5 metabolites



parcel_id	parcel_name	hemisphere	anatomical_coverage_percent	mean_crlb	qc_valid_fraction
2	Left-Cerebral-White-Matter	L	100.000	8.810	0.959
7	Left-Cerebellum-White-Matter	L	100.000	12.762	0.296
8	Left-Cerebellum-Cortex	L	100.000	17.476	0.418
10	Left-Thalamus	L	100.000	7.808	0.992
11	Left-Caudate	L	100.000	9.278	0.928
12	Left-Putamen	L	100.000	7.541	0.968
13	Left-Pallidum	L	100.000	10.689	0.978
16	Brain-Stem	NaN	100.000	16.395	0.546

parcel_id	parcel_name	hemisphere	anatomical_coverage_percent	mean_crlb	qc_valid_fraction
17	Left-Hippocampus	L	100.000	13.014	0.850
18	Left-Amygdala	L	100.000	21.233	0.317
26	Left-Accumbens-area	L	100.000	14.080	0.720
28	Left-VentralDC	L	100.000	11.766	0.945
41	Right-Cerebral-White-Matter	R	100.000	7.669	0.978
46	Right-Cerebellum-White-Matter	R	100.000	17.312	0.222
47	Right-Cerebellum-Cortex	R	100.000	30.546	0.398
49	Right-Thalamus	R	100.000	6.915	0.997
50	Right-Caudate	R	100.000	9.826	0.911
51	Right-Putamen	R	100.000	7.628	0.978
52	Right-Pallidum	R	100.000	9.740	0.960
53	Right-Hippocampus	R	100.000	11.088	0.904
54	Right-Amygdala	R	100.000	12.180	0.800
58	Right-Accumbens-area	R	100.000	17.080	0.800
60	Right-VentralDC	R	100.000	10.852	0.871
1001	ctx-lh-bankssts	L	100.000	5.729	1.000
1002	ctx-lh-caudalanteriorcingulate	L	100.000	9.718	0.955
1003	ctx-lh-caudalmiddlefrontal	L	100.000	6.670	0.793
1005	ctx-lh-cuneus	L	100.000	6.842	1.000
1006	ctx-lh-entorhinal	L	100.000	113.975	0.312
1007	ctx-lh-fusiform	L	100.000	28.516	0.691
1008	ctx-lh-inferiorparietal	L	100.000	7.126	0.903
1009	ctx-lh-inferiortemporal	L	100.000	31.210	0.429
1010	ctx-lh-isthmuscingulate	L	100.000	5.710	1.000
1011	ctx-lh-lateraloccipital	L	100.000	9.049	0.476
1012	ctx-lh-lateralorbitofrontal	L	100.000	22.441	0.785
1013	ctx-lh-lingual	L	100.000	6.737	0.968
1014	ctx-lh-medialorbitofrontal	L	100.000	33.200	0.836
1015	ctx-lh-middletemporal	L	100.000	22.430	0.766
1016	ctx-lh-parahippocampal	L	100.000	11.832	0.884
1017	ctx-lh-paracentral	L	100.000	6.874	0.995
1018	ctx-lh-parsopercularis	L	100.000	6.584	1.000
1019	ctx-lh-parsorbitalis	L	100.000	7.612	0.941
1020	ctx-lh-parstriangularis	L	100.000	8.074	0.966
1021	ctx-lh-pericalcarine	L	100.000	7.240	0.960
1022	ctx-lh-postcentral	L	100.000	6.703	0.837
1023	ctx-lh-posteriorcingulate	L	100.000	5.994	1.000

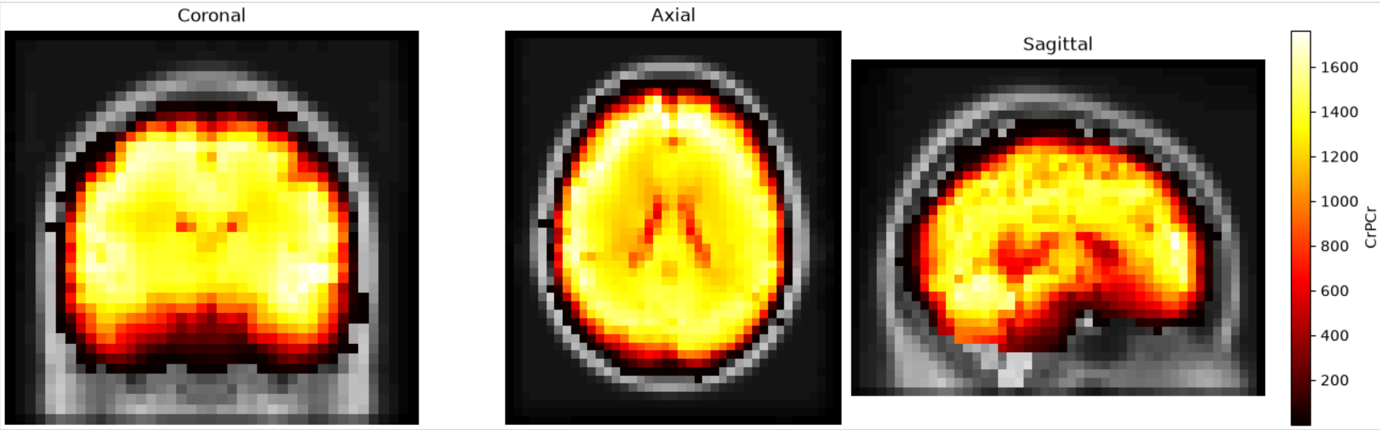
parcel_id	parcel_name	hemisphere	anatomical_coverage_percent	mean_crlb	qc_valid_fraction
1024	ctx-lh-precentral	L	100.000	6.887	0.933
1025	ctx-lh-precuneus	L	100.000	6.252	0.987
1026	ctx-lh-rostralanteriorcingulate	L	100.000	17.752	0.920
1027	ctx-lh-rostralmiddlefrontal	L	100.000	8.278	0.586
1028	ctx-lh-superiorfrontal	L	100.000	8.015	0.849
1029	ctx-lh-superiorparietal	L	100.000	7.070	0.922
1030	ctx-lh-superiortemporal	L	100.000	6.975	0.961
1031	ctx-lh-supramarginal	L	100.000	6.670	0.863
1032	ctx-lh-frontalpole	L	100.000	20.200	0.325
1033	ctx-lh-temporalpole	L	100.000	33.556	0.478
1034	ctx-lh-transversetemporal	L	100.000	5.860	1.000
1035	ctx-lh-insula	L	100.000	7.332	0.971
2001	ctx-rh-bankssts	R	100.000	6.783	1.000
2002	ctx-rh-caudalanteriorcingulate	R	100.000	15.320	0.790
2003	ctx-rh-caudalmiddlefrontal	R	100.000	7.593	0.900
2005	ctx-rh-cuneus	R	100.000	6.712	0.971
2006	ctx-rh-entorhinal	R	100.000	43.578	0.222
2007	ctx-rh-fusiform	R	100.000	17.551	0.841
2008	ctx-rh-inferiorparietal	R	100.000	6.393	0.966
2009	ctx-rh-inferiortemporal	R	100.000	41.921	0.687
2010	ctx-rh-isthmuscingulate	R	100.000	5.991	1.000
2011	ctx-rh-lateraloccipital	R	100.000	7.697	0.623
2012	ctx-rh-lateralorbitofrontal	R	100.000	25.413	0.770
2013	ctx-rh-lingual	R	100.000	6.972	0.997
2014	ctx-rh-medialorbitofrontal	R	100.000	14.895	0.943
2015	ctx-rh-middletemporal	R	100.000	12.075	0.916
2016	ctx-rh-parahippocampal	R	100.000	9.262	0.975
2017	ctx-rh-paracentral	R	100.000	6.640	1.000
2018	ctx-rh-parsopercularis	R	100.000	7.507	1.000
2019	ctx-rh-parsorbitalis	R	100.000	41.329	0.412
2020	ctx-rh-parstriangularis	R	100.000	9.600	0.940
2021	ctx-rh-pericalcarine	R	100.000	6.287	1.000
2022	ctx-rh-postcentral	R	100.000	6.628	0.879
2023	ctx-rh-posteriorcingulate	R	100.000	5.890	1.000
2024	ctx-rh-precentral	R	100.000	6.896	0.973
2025	ctx-rh-precuneus	R	100.000	5.825	1.000
2026	ctx-rh-rostralanteriorcingulate	R	100.000	12.752	0.895

parcel_id	parcel_name	hemisphere	anatomical_coverage_percent	mean_crlb	qc_valid_fraction
2027	ctx-rh-rostralmiddlefrontal	R	100.000	19.033	0.674
2028	ctx-rh-superiorfrontal	R	100.000	7.344	0.934
2029	ctx-rh-superiorparietal	R	100.000	7.076	0.927
2030	ctx-rh-superiortemporal	R	100.000	6.728	0.998
2031	ctx-rh-supramarginal	R	100.000	6.075	0.928
2032	ctx-rh-frontalpole	R	100.000	332.378	0.140
2033	ctx-rh-temporalpole	R	100.000	172.214	0.300
2034	ctx-rh-transversetemporal	R	100.000	5.508	1.000
2035	ctx-rh-insula	R	100.000	7.436	0.945

MNI-space alignment

MNI-space alignment

Reference metabolite map (`CrPCr`) overlaid on the full-head MNI152 template, opaque so placement inside the brain can be verified post-alignment.



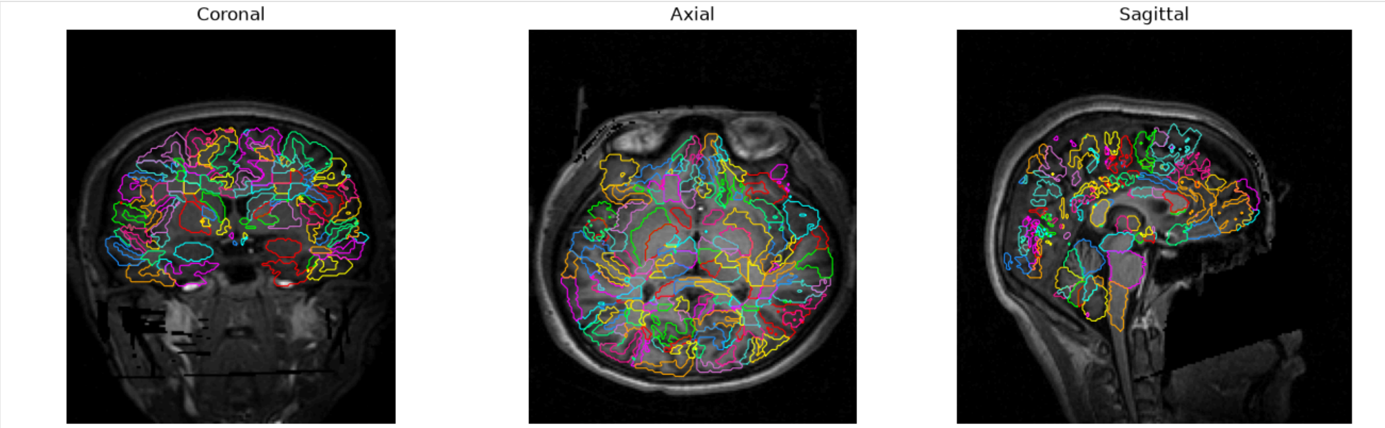
Signal-weighted leakage

metabolite	n_quality_voxels	total_signal_mass	leakage_signal_mass	leakage_fraction	leakage_percent
CrPCr	75162	15068980.000	113446.664	0.008	0.753
GPCPCh	75308	4208737.000	36944.441	0.009	0.878
GluGln	75100	20511016.000	185461.547	0.009	0.904
Ins	74806	12791346.000	108439.250	0.008	0.848
NAANAAG	75756	23747760.000	183141.062	0.008	0.771

Parcellation

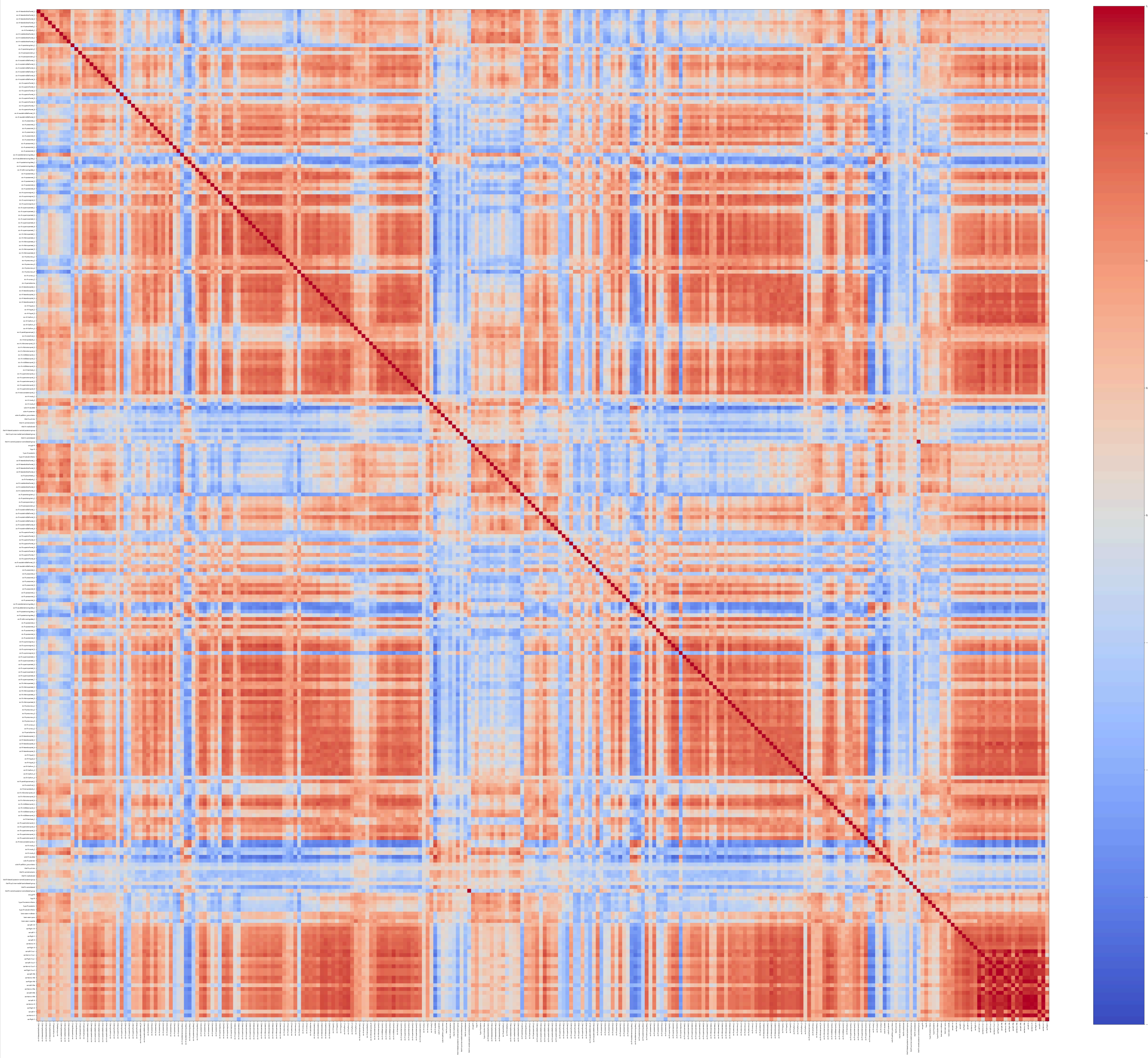
Parcellation outlines (T1w space)

276 regions.



Connectivity

Connectivity matrix (spearman)



Runtime

Per-step duration

nproc: `1` | nthreads: `16`

Excludes this report-generation step's own duration (not yet known while it's still running) and any time spent before this recording's pipeline started (e.g. queued behind other recordings under `--nproc`).

Step	Duration	% of total
Tissue segmentation	41.1s	67.7%
Anatomical preparation	0.3s	0.5%
MRSI preprocessing	4.1s	6.8%
MRSI-T1w-MNI registration	0.2s	0.3%
Tissue probability maps in MRSI space	0.9s	1.5%
Partial volume correction	0.2s	0.3%
Resampling MRSI maps to T1w/MNI space	3.7s	6.0%
Signal leakage QC	0.0s	0.1%
SynthSeg parcellation and QC	1.9s	3.2%
Parcellation	2.6s	4.3%
Regional metabolite extraction	2.2s	3.5%
Regional metabolic profiles and connectivity	3.5s	5.8%
Total (through report generation)	**1m 00.8s**	100.0%

Outputs

- **atlas_mrsi**: `/data/derivatives/mrsiprep/sub-01/ses-01/anat/synthseg/sub-01_ses-01_space-mrsi_atlas-chimeraLFMIHIFIS_scale3_dseg.nii.gz`
- **connectivity**: `{'profiles': PosixPath('/data/derivatives/mrsiprep/sub-01/ses-01/connectivity/sub-01_ses-01_atlas-chimeraLFMIHIFIS_scale3_npert-50_filt-biharmonic_pvcorr_GM_desc-metabolicprofiles_mrsi.npz'), 'matrix_npz': PosixPath('/data/derivatives/mrsiprep/sub-01/ses-01/connectivity/sub-01_ses-01_atlas-chimeraLFMIHIFIS_scale3_npert-50_filt-biharmonic_pvcorr_GM_desc-connectivity_mrsi.npz'), 'matrix_tsv': PosixPath('/data/derivatives/mrsiprep/sub-01/ses-01/connectivity/sub-01_ses-01_atlas-chimeraLFMIHIFIS_scale3_npert-50_filt-biharmonic_pvcorr_GM_desc-connectivity_mrsi.tsv'), 'nodes': PosixPath('/data/derivatives/mrsiprep/sub-01/ses-01/connectivity/sub-01_ses-01_atlas-chimeraLFMIHIFIS_scale3_npert-50_filt-biharmonic_pvcorr_GM_desc-nodes_mrsi.tsv'), 'edges': PosixPath('/data/derivatives/mrsiprep/sub-01/ses-01/connectivity/sub-01_ses-01_atlas-chimeraLFMIHIFIS_scale3_npert-50_filt-biharmonic_pvcorr_GM_desc-edges_mrsi.tsv')}`
- **leakage_qc**: `/data/derivatives/mrsiprep/sub-01/ses-01/confounds/sub-01_ses-01_desc-leakageqc.tsv`
- **metprofiles**: `/data/derivatives/mrsiprep/sub-01/ses-01/mrsi/parcel/sub-01_ses-01_atlas-chimeraLFMIHIFIS_scale3_desc-GMmetprofiles_mrsi.npz`
- **mrsi_reference**: `/data/derivatives/mrsiprep/sub-01/ses-01/mrsi/orig/sub-01_ses-01_space-mrsi_desc-reference_mrsi.nii.gz`
- **parcel_qc**: `/data/derivatives/mrsiprep/sub-01/ses-01/anat/synthseg/sub-01_ses-01_atlas-synthseg_desc-parcelqc.tsv`
- **preliminary_atlas_mrsi**: `/data/derivatives/mrsiprep/sub-01/ses-01/anat/synthseg/sub-01_ses-01_space-mrsi_atlas-synthseg_desc-robustGMWM_dseg.nii.gz`
- **qc_summary**: `/data/derivatives/mrsiprep/sub-01/ses-01/confounds/sub-01_ses-01_desc-mrsiqc.tsv`
- **regional_table**: `/data/derivatives/mrsiprep/sub-01/ses-01/anat/synthseg/sub-01_ses-01_space-mrsi_atlas-chimeraLFMIHIFIS_scale3_desc-regional_metabolites.tsv`
- **regional_tissue_regression**: `None`
- **registration_t1w**: `/data/derivatives/mrsiprep/sub-01/ses-01/anat/synthseg/sub-01_ses-01_space-T1w_desc-brainCSF_T1w.nii.gz`
- **t1w**: `/data/derivatives/mrsiprep/sub-01/ses-01/anat/synthseg/sub-01_ses-01_space-T1w_desc-synthsegBrain_T1w.nii.gz`
- **tissue_4d**: `/data/derivatives/work/sub-01/ses-01/pvc/sub-01_desc-4Dtissue_mrsi.nii.gz`
- **transformed_maps**: `{'MNI152NLin2009cAsym': {'CrPCr': PosixPath('/data/derivatives/mrsiprep/sub-01/ses-01/mrsi/mni/sub-01_ses-01_space-MNI152NLin2009cAsym_res-5_met-CrPCr_desc-signal_mrsi.nii.gz'), 'GluGln': PosixPath('/data/derivatives/mrsiprep/sub-01/ses-01/mrsi/mni/sub-01_ses-01_space-MNI152NLin2009cAsym_res-5_met-GluGln_desc-signal_mrsi.nii.gz'), 'GPCPCh': PosixPath('/data/derivatives/mrsiprep/sub-01/ses-01/mrsi/mni/sub-01_ses-01_space-MNI152NLin2009cAsym_res-5_met-GPCPCh_desc-signal_mrsi.nii.gz'), 'NAANAAG': PosixPath('/data/derivatives/mrsiprep/sub-01/ses-01/mrsi/mni/sub-01_ses-01_space-MNI152NLin2009cAsym_res-5_met-NAANAAG_desc-signal_mrsi.nii.gz'), 'Ins': PosixPath('/data/derivatives/mrsiprep/sub-01/ses-01/mrsi/mni/sub-01_ses-01_space-MNI152NLin2009cAsym_res-5_met-Ins_desc-signal_mrsi.nii.gz'), 'crlb-CrPCr': PosixPath('/data/derivatives/mrsiprep/sub-01/ses-01/confounds/sub-01_ses-01_space-MNI152NLin2009cAsym_res-5_met-CrPCr_desc-crlb_mrsi.nii.gz'), 'crlb-GluGln': PosixPath('/data/derivatives/mrsiprep/sub-01/ses-01/confounds/sub-01_ses-01_space-MNI152NLin2009cAsym_res-5_met-GluGln_desc-crlb_mrsi.nii.gz'), 'crlb-GPCPCh': PosixPath('/data/derivatives/mrsiprep/sub-01/ses-01/confounds/sub-01_ses-01_space-MNI152NLin2009cAsym_res-5_met-GPCPCh_desc-crlb_mrsi.nii.gz'), 'crlb-NAANAAG': PosixPath('/data/derivatives/mrsiprep/sub-01/ses-01/confounds/sub-01_ses-01_space-MNI152NLin2009cAsym_res-5_met-NAANAAG_desc-crlb_mrsi.nii.gz'), 'crlb-Ins': PosixPath('/data/derivatives/mrsiprep/sub-01/ses-01/confounds/sub-01_ses-01_space-MNI152NLin2009cAsym_res-5_met-Ins_desc-crlb_mrsi.nii.gz'), 'snr': PosixPath('/data/derivatives/mrsiprep/sub-01/ses-01/confounds/sub-01_ses-01_space-MNI152NLin2009cAsym_res-5_desc-snr_mrsi.nii.gz'), 'fwhm': PosixPath('/data/derivatives/mrsiprep/sub-01/ses-01/confounds/sub-01_ses-01_space-MNI152NLin2009cAsym_res-5_desc-fwhm_mrsi.nii.gz')}}`

Citations

MRSIPrep: see `CITATION.cff` in the [MRSIPrep repository] for how to cite this software.

MRSI acquisition reporting follows the MRSinMRS minimum reporting standard (Lin et al. 2021).